



COLLEGE OF ARTS & SCIENCES

BACHELOR OF SCIENCE IN ENVIRONMENTAL SCIENCE

Effective AY 2023 - 2024

Based on CMO # 35, series of 2005

BOR Resolution # _____

Academic Year 2023 - 2024

FIRST YEAR							
First Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
BIO 1	General Biology I	3	2	3	6	NONE	
GE MMW	Mathematics in the Modern World	3	0	3	0	NONE	
GE CD	Communication Development	3	0	3	0	NONE	
GE US	Understanding the Self	3	0	3	0	NONE	
ES 101	Introduction to Environmental Science	3	0	3	0	NONE	
PATHFit 1	Movement Competency Training	2	0	2	0	NONE	
NSTP 1	ROTC or CWTS	3	0	3	0	NONE	
TOTAL		22					

FIRST YEAR							
Second Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
BIO 2	General Biology II	3	2	3	6	BIO 1	
CALC 1	Calculus I	4	0	4	0	NONE	
GE PC	Purposive Communication	3	0	3	0	NONE	
ES 102	Ecosystem, Structure and Dynamics	3	0	3	0	ES 101	
SS 1	People, Culture and Environmental Issues in Caraga	3	0	3	0	NONE	
PATHFit 2	Exercise - Based Fitness Activities	2	0	2	0	PATHFit 1	
NSTP 2	ROTC or CWTS	3	0	3	0	NSTP 1	
TOTAL		23					

SECOND YEAR							
First Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
BIO 109	General Ecology	3	2	3	6	BIO 2	
CALC 2	Calculus II	4	0	4	0	CALC 1	
GE TW	Technical Writing	3	0	3	0	NONE	
SCI 1	Meteorology	3	0	3	0	NONE	
GE IT	Information Technology	2	1	2	3	NONE	
EST 101	Environmental Laws and Policy	3	0	3	0	NONE	
PATHFit 3	Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities	2	0	2	0	PATHFit 2	
TOTAL		23					

SECOND YEAR							
Second Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
CHEM 100	General Chemistry 1 (Inorganic)	3	2	3	6	NONE	
STAT 1	Environmental Statistics	3	1	3	3	NONE	
GE STS	Science and Technology and Society	3	0	3	0	NONE	
GE RPH	Readings in Philippine History	3	0	3	0	NONE	
ES 103	Geographic Information Systems and Remote Sensing	3	0	3	0	GE IT	
ES 104	Landscape Ecology	3	0	3	0	BIO 109	
PATHFit 4	Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities	2	0	2	0	PATHFit 3	
TOTAL		23					

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THIRD YEAR							
First Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
CHEM 101	General Chemistry 2 (Organic)	3	2	3	6	Chem 100	
PHYS 1	Mechanics	2	1	2	3	Calc 1	
GE TCW	The Contemporary World	3	0	3	0	NONE	
GE E	Ethics	3	0	3	0	NONE	
ES 105	Aquatic Ecology	3	0	3	0	BIO 109	
EST 102	Climate Change Adaptation and Disaster Risk Reduction	3	0	3	0	NONE	
EST 103	Community Resource Based Management	3	0	3	0	SS 1	
TOTAL		23					

THIRD YEAR							
Second Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
CHEM 103	Analytical Chemistry	3	2	3	6	Chem101	
ES 1	Earth Science	3	2	3	6	NONE	
SCI 2	Geography	3	0	3	0	NONE	
PHYS 2	Thermodynamics	2	1	2	3	PHYS 1, CALC 1	
GE RL	Life and Works of Rizal	3	0	3	0	NONE	
EST 104	Coastal Resource Management	3	0	3	0	ES 105	
TOTAL		22					

THIRD YEAR							
Summer		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
ELEC 1	Indutries Inspection and Seminar	0	1	0	3	at least 32 units of ES major	
OJT	On the Job Training	3	0		250		
TOTAL		4					

FOURTH YEAR							
First Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
RES 1	Methods of Environmental Research (Thesis Proposal)	3	0	3	0	STAT 1	
GE WL	World Literature	3	0	3	0	NONE	
GE AA	Art Appreciation	3	0	3	0	NONE	
ES 106	Natural Resource Economics and Val	3	0	3	0	STAT 1	
TOTAL		12					

FOURTH YEAR							
Second Semester		UNITS		HOURS			
Course Code	Course Title	LEC	LAB	LEC	LAB	PRE-REQUISITE	GRADE
Res 2	Thesis (Final Defense)	3	0	3	0	RES 1	
EIA	Environmental Impact Assessment	3	0	3	0	ALL MAJOR SUBJECTS	
ES 107	Environmental Monitoring and Evaluation	3	2	3	6	ALL MAJOR SUBJECTS	
TOTAL		11					

*Every Laboratory unit shall be conducted in three (3) hours pursuant to CMO 40, series of 2008-Manual of Operation of Private Higher Education (MORPHE) supplemented by CMO 30, series of 2009 as to its applicability of State Universities and Colleges (SUCs).

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COMPONENTS	CMO Requirements	This Curriculum
General Education	36	36
Core Science	52	53
Biology	15	15
Chemistry	15	15
Earth Science	5	5
Mathematics	12	12
Physics	5	6
Environmental Science with Environmental Monitoring	26	26
Environmental Studies/ Environmental Management	12	12
EIA System	3	3
Thesis or Special Project+ OJT	6	9
Institutional		10
Elective		
PATHFit	8	8
NSTP	6	6
TOTAL	149	163

SUMMARY	1	2	SUMMER	TOTAL
1st year	22	23		45
2nd year	23	23		46
3rd year	23	22	4	49
4th year	12	11		23
TOTAL	80	79	4	163

Prepared by:

DORIE T. GEDUCOS, MST
BSES, Program Coordinator

JOCELYN B. PANDIYOS, EdD
Department Chair, DMNS

Reviewed by:

ERWIN B. BERRY, EdD
Curriculum Director and Development

Recommending Approval:

ROMEO A. PATAN, EdD
Dean, College of Arts & Sciences

Approved:

MA. LADY SOL A. SUAZO, PhD
Vice President for Academic Affairs

Reviewed and noted by:

CHED Caraga Regional Supervisor

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SUMMARY			
School Offering			Units
A. General Education Courses (36 units)			36
1	GE US	Understanding the Self	3
2	GE RPH	Readings in Philippine History	3
3	GE CW	The Contemporary World	3
4	GE MMW	Mathematics in the Modern World	3
5	GE PC	Purposive Communication	3
6	GE AA	Art Appreciation	3
7	GE STS	Science, Technology and Society	3
8	GE E	Ethics	3
9	GE RIZAL	Life and Works Rizal	3
10	GE WL	World Literature	3
11	GE IT	Information Technology	3
12	SS 1	People, Culture, and Environment Issues in Caraga	3
B. Mandated Courses (14 units)			8
1	PATHFIT 1	PATHFit 1- Movement Competency Training or MCT	2
2	PATHFIT 2	PATHFit 2- Exercise-based Fitness Activities	2
3	PATHFIT 3	PATHFit 3- Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities	2
4	PATHFIT 4	PATHFit 4- Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities	2
NSTP			6
5	NSTP 1	National Service Training Program 1	3
6	NSTP 2	National Service Training Program 2	3
Core Subjects			53
Biology			15
1	BIO 1	General Biology I	5
2	BIO 2	General Biology II	5
3	BIO 109	General Ecology	5
Chemistry			15
1	CHEM 100	General Chemistry 1 (Inorganic)	5
2	CHEM 101	General Chemistry II (Organic)	5
3	CHEM 103	Analytical Chemistry	5
Earth Science			5
1	ES 1	Earth Science	5
Mathematics			12
1	CALC 1	Calculus I	4
2	CALC 2	Calculus II	4
3	STAT 1	Environmental Statistics	4
Physics			6
1	PHYS 1	Mechanics	3
2	PHYS 2	Thermodynamics	3
D. Environmental Science with Environmental Monitoring (lec/lab)			26
1	ES 101	Introduction to Environmental Science	3
2	ES 102	Ecosystem, Structure, and Dynamics	3
3	ES 103	Geographic Information Systems and Remote Sensing	3
4	ES 104	Landscape Ecology	3
5	ES 105	Aquatic Ecology	3
6	ES 107	Environmental Monitoring and Evaluation	5
7	SCI 1	Meteorology	3
8	SCI 2	Geography	3

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E. Environmental Studies/ Environmental Management			12
1	EST 101	Environmental Laws and Policy	3
2	EST 102	Climate Change Adaptation and Disaster Risk Reduction	3
3	EST 103	Community Resource Based Management	3
4	EST 104	Coastal Resource Management	3
5	EIA	Environmental Impact Assessment	3
D. Institutional Subjects			10
1	GE CD	Communication Development	3
2	GE TW	Technical Writing	3
3	ELEC 1	Industries Inspection and Seminar (Elective)	1
4	ES 106	Natural Resources Economics and Valuation	3
E. Thesis/Practicum			9
1	OJT	On the Job Training (3 Units)	3
2	RES 1	Methods of Environmental Research (Thesis Proposal)	3
3	RES 2	Thesis (Final Thesis)	3

		NEMSU Curriculum		3
		SUMMARY		
A.		General Education Courses (36 units)	Units	
B.		Mandated Courses (14 units) (PATHfit and NSTP)		36
C.		Core Subjects (52 Units)		14
		Biology		53
		Chemistry		15
		Earth Science		15
		Mathematics		5
		Physics		12
		Environmental Science with Environmental Monitoring (lec/lab)		6
		Environmental Studies/ Environmental Management		26
		The IEA System		12
D.		Thesis/Practicum (6 Units)		3
F.		Institutional Subjects		9
				10
		Total		163

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